

Exhaustive Proof Some theorems can be proved by examining a relatively small number of examples. Such proofs are called **Exhaustive proofs**, or **proofs by exhaustion** because these proofs proceed by examining all possibilities. An exhaustive proof is a special type of proof by cases where each case involves checking a single example.

Example 1: Prove that $(n+1)^3 \gg 3^n$ if n is a positive integer with $n \leq 4$.

Solution: Let $P(n)$ be $(n+1)^3 \gg 3^n$.
 We need to prove that $P(0) \vee P(1) \vee P(2) \vee P(3) \rightarrow$ ~~$P(0) \vee P(1) \vee P(2) \vee P(3)$~~ is True

$\therefore$ We need to show that each of $P(0)$

Since $n \leq 4$, we need to prove that each of $P(0), P(1), P(2)$ and $P(3)$ are True.

$P(0)$: $(0+1)^3 = 1 \gg 3^0 = 1$. $P(0)$ is True

$P(1)$: LHS = $(1+1)^3 = 8$ RHS = $3^1 = 3$. $8 \gg 3$. $\therefore P(1)$ is True

$P(2)$: LHS = $(2+1)^3 = 27$ RHS = $3^2 = 9$. $27 \gg 9$. $\therefore P(2)$ is True

$P(3)$: LHS = $(3+1)^3 = 64$ RHS = $3^3 = 27$. $64 \gg 27$. $\therefore P(3)$ is True.

We have used the method of exhaustion to prove that $(n+1)^3 \gg 3^n$ if n is a positive integer with $n \leq 4$. $\square$

$\rightarrow P(4)$: LHS = $(4+1)^3 = 125$ RHS = $3^4 = 81$. $125 \gg 81$. $\therefore P(4)$ is True

Example 2: Prove that the only consecutive positive integers not exceeding 100 that are perfect powers are 8 and 9. (An integer n is a

Perfect power if it equals m^a , where m is an integer and a is an integer greater than 1.)